INCH-POUND

MIL-C-3098/77C
28 June 1995
SUPERSEDING
MIL-C-3098/77B
23 June 1975

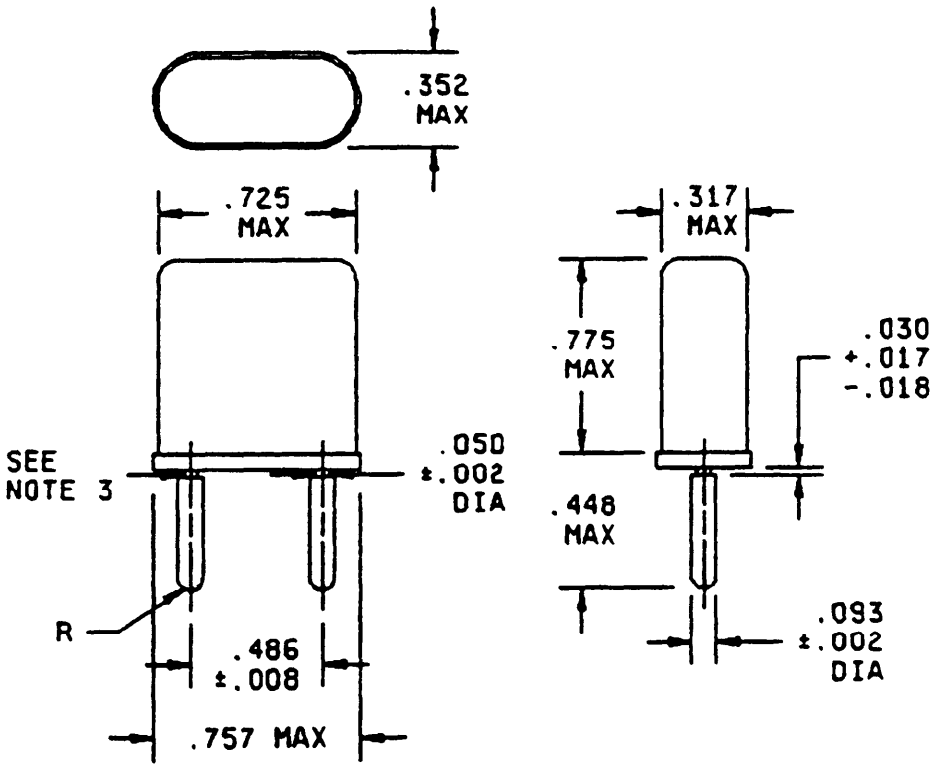
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR5/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 2 to 10 MHz; fundamental; noncontrolled; antiresonance.



Inches	mm
.002	0.05
.008	0.20
.017	0.43
.018	0.46
.030	0.76
.050	1.27
.093	2.36
.317	8.05
.352	8.94
.448	11.38
.486	12.34
.725	18.42
.757	19.23
.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR5/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS

Holder: HC-17/U (MIL-H-10056/8).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.0 to 10.0 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 2.0 to 5.1 MHz - 50.0 ±5.0 mW.
5.1+ to 10.0 MHz - 30.0 ±3.0 mW.

Operating temperature range: -55°C to +90°C, inclusive.

Frequency tolerance: ±50 ppm.

Equivalent resistance: See table I.

Antiresonance, load capacitance: 25.0 ±0.5 pF.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

TABLE I. Equivalent resistance.

Frequency range, inclusive MHz	Maximum resistance Ohms
2.0 to 2.1	550
2.1+ to 2.225	500
2.225+ to 2.3	450
2.3+ to 2.45	400
2.45+ to 2.6	350
2.6+ to 2.8	300
2.8+ to 3.05	250
3.05+ to 3.4	200
3.4+ to 3.6	180
3.6+ to 3.9	150
3.9+ to 4.25	130
4.25+ to 4.65	100
4.65+ to 5.1	75
5.1+ to 6.5	50
6.5+ to 7.5	37
7.5+ to 9.0	25
9.0+ to 10.0	20

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Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

NOTES:
1. Socket. This unit has oversize pins and does not use same socket as preferred types of crystal units.
2. This crystal-unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0680-18)

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